

Representing & Interpreting Data: Notes & Graphic Organizer

COMMON CORE STANDARDS:

CCSS.HSS.ID.A.1 Represent data with plots on the real number line (dot plots, histograms, and box plots).

CCSS.HSS.ID.A.3 Interpret differences in shape, center, and spread in the context of the data sets, accounting for possible effects of extreme data points (outliers).

SKILLS USED:

Creating dot plots, histograms, box plots, and interpreting shape of data.

DIRECTIONS:

This resource is used in conjunction with the Representing and Interpreting Data Power Point and "Interpreting the Distribution of Data" Handout. Students follow along with the Cornell style notes. Then create a dot plot, histogram, and box plot in the examples. Be prepared to discuss or have a step-by-step on how to use the STATS function of the calculator to do the examples (I didn't include this in the PowerPoint because every classroom uses different calculators.) However, the Power Point is EDITABLE, so please feel free to add calculator instructions for your class ☺ For the summary, let students summarize their own notes, have them answer the essential question, or let them pick a question to answer from the last slide of the presentation of the Power Point, your choice! After reviewing the "Interpreting the Distribution of Data" Handout, go back and discuss the shapes of the examples.

DIFFERENTIATION:

Two tiers of differentiation are provided in this Cornell Notes resource. One set of notes has the notes filled in, and for the other set of notes, students will have to write in their notes. On both sets of notes, students will have to work out the example problems.

Representing and Interpreting Data



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Dot Plot	- Graphic display of data using dots. - The data values are on the horizontal axis. - The frequency of each data is represented by dots.	Ex: A survey for "How many green shirts do you own?" had these results: 5, 3, 3, 0, 6, 2, 1, 4, 3, 7, 2, 4, 5, 4, 2, 5, 1, 6, 0 Shape:
Histogram Note: Histograms are different from Bar Graphs. Histograms have no gaps, and measure continuous data like numbers.	- Graphic display of data using bars - The data values are grouped in number ranges, or class intervals, on the horizontal axis - The frequency of each data is on the vertical axis (height of bar)	Ex: A survey for the heights (in inches) of students in had these results: 51, 53, 68, 74, 66, 54, 55, 57, 72, 56, 59, 60, 61, 63 Shape:
Box plot aka Box and Whiskers aka 5 Number Summary	- Find the 5 number summary of the data set: min., Q1, median, Q3, max. - Create a horizontal or vertical number line, make sure to use a scale that includes the numbers from your 5 number summary - Use the 5 numbers to create a box plot.	Ex: A survey for the average miles per gallon(mpg) of each car in a parking lot had these results: 29, 22, 13, 28, 24, 32, 25, 28, 27, 22, 30 Shape:

Summary:

Representing and Interpreting Data



Common Core Standards:

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Dot Plot		Ex: A survey for "How many green shirts do you own?" had these results: 5, 3, 3, 0, 6, 2, 1, 4, 3, 7, 2, 4, 5, 4, 2, 5, 1, 6, 0 Shape:
Histogram Note: Histograms are different from Bar Graphs. Histograms have no gaps, and measure continuous data like numbers.		Ex: A survey for the heights (in inches) of students in had these results: 51, 53, 68, 74, 66, 54, 55, 57, 72, 56, 59, 60, 61, 63 Shape:
Box plot aka Box and Whiskers aka 5 Number Summary		Ex: A survey for the average miles per gallon(mpg) of each car in a parking lot had these results: 29, 22, 13, 28, 24, 32, 25, 28, 27, 22, 30 Shape:

Summary:

Interpreting the Distribution of Data

Common Core Standards:

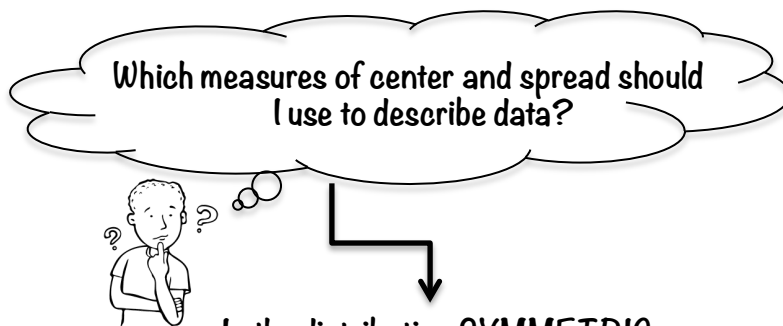
CCSS.HSS.ID.A.1 Represent data with plots on the real number line (dot plots, histograms, and box plots).

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The distribution of data shows how the data is arranged. We can discuss distribution in 3 ways:

- 1) Center
- 2) Spread (variation)
- 3) Shape



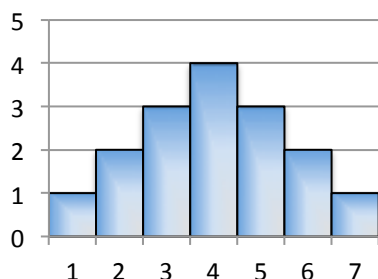
Is the distribution **SYMMETRIC**
(left side looks like right side)?

YES

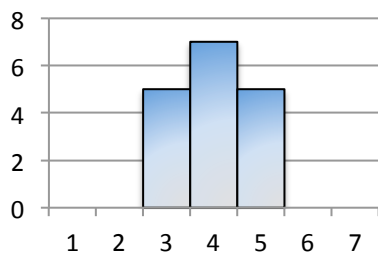
NO

Use the mean to discuss the center and use the mean absolute deviation to discuss spread.

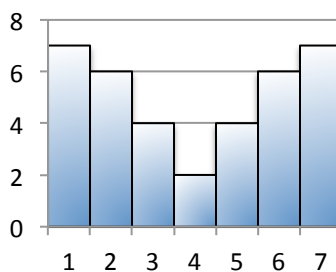
Use the median to discuss the center and use the interquartile range to discuss the spread.



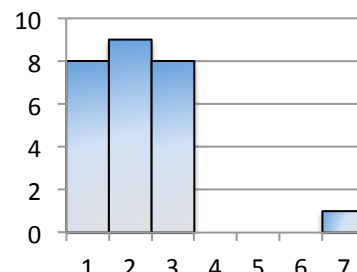
Normal/ Bell Shape Distribution,
Symmetric, More Spread



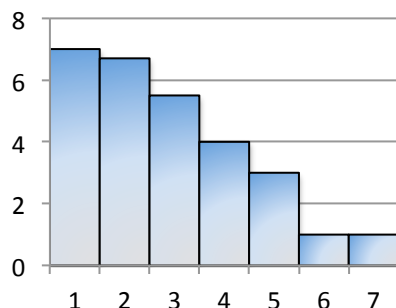
Unimodal, Symmetric, Less Spread, mean=median



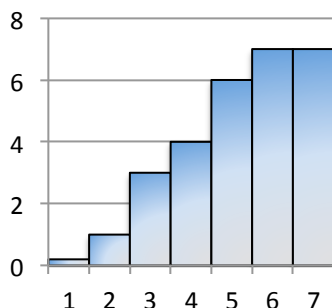
U-Shape Distribution,
Symmetric, Bimodal



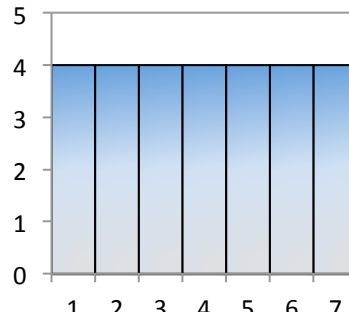
Asymmetric, Gap, Outlier



Skewed Right,
Positive Skew,
mean>median



Skewed Left,
Negative Skew,
median>mean



Uniform

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Jennifer Byrd

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